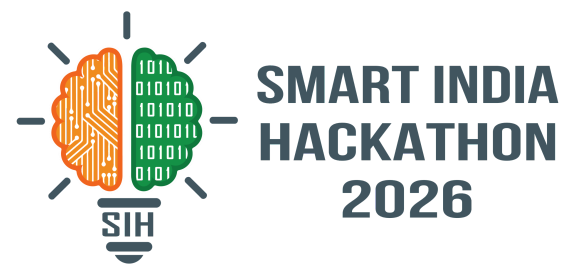


SMART INDIA HACKATHON 2026



- **Problem Statement ID :- SIH26167**
- **Problem Statement Title :- SatQuery AI - An Interactive Vision-Language Assistant for Multimodal Remote Sensing Image Analysis through Text Queries**
- **Theme :- Space Technology**
- **PS Category :- Software**
- **Team ID -**
- **Team Name :- SouL**



Problem

- Existing satellite image analysis systems:
- Require GIS and remote sensing expertise
- Are designed for one predefined task only
- Cannot understand natural language
- Cannot jointly analyze
Optical + SAR + Multi-temporal imagery
- Provide little explainability to non-expert users

Innovation & Uniqueness

- Agentic AI-based workflow orchestration
- Automatic task routing
- Optical + SAR fusion reasoning
- Multi-temporal change analysis
- Domain-adapted Vision-Language Models
- Explainable AI with confidence score & execution summary

Key Features

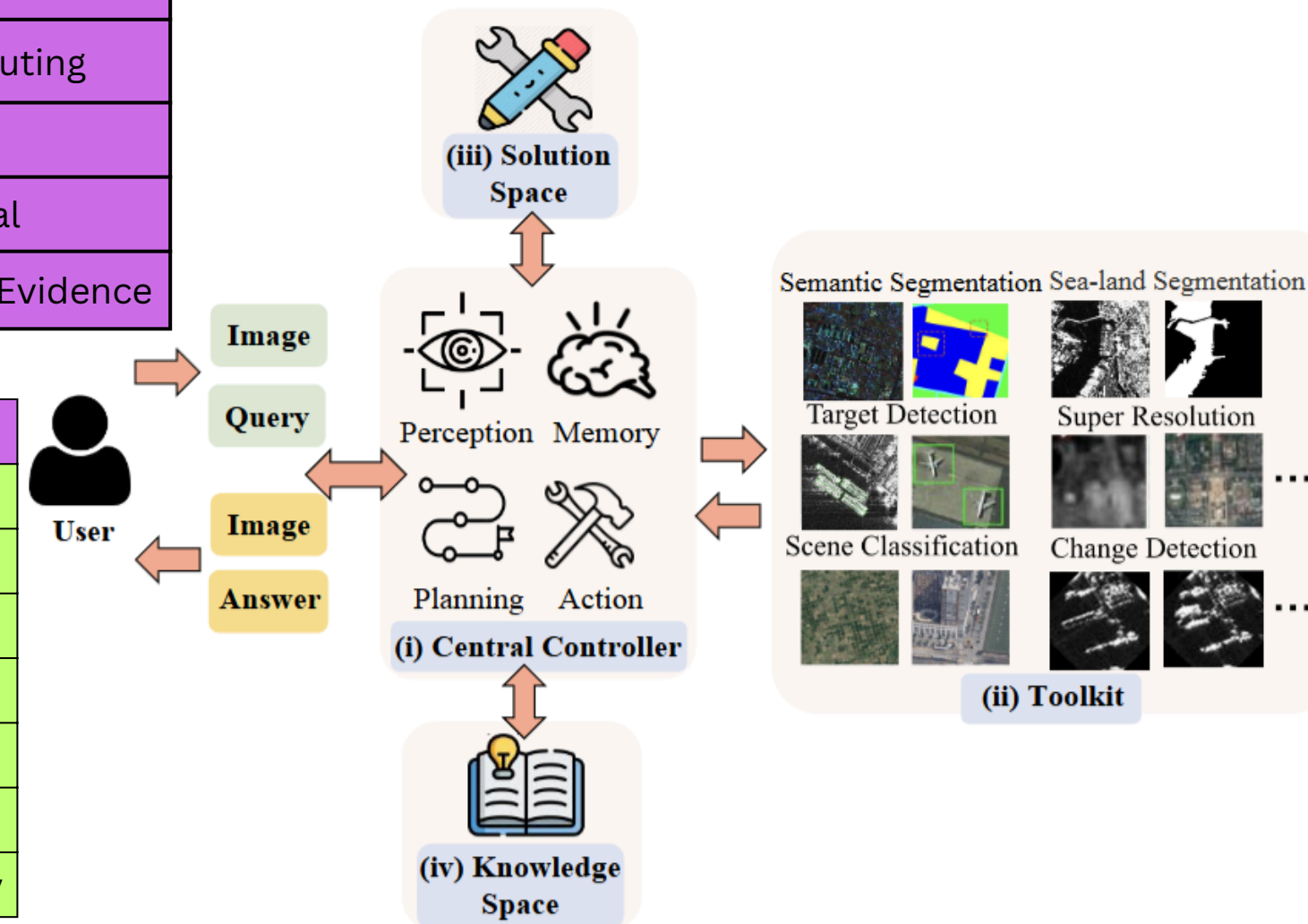
- Multimodal Image Understanding
- Agentic AI Orchestration
- Natural Language Queries
- Optical + SAR Fusion
- Multi-temporal Change Detection
- Confidence Score
- Visual Grounding + Downloadable reports

Proposed Solution

- SatQuery AI is an Agentic Vision-Language Assistant that enables users to analyze satellite imagery using natural-language queries.
- The system automatically:
 - Understands the user's query
 - Validates uploaded satellite images
 - Identifies the required remote sensing task
 - Routes the request to specialized AI models
 - Combines outputs into a single evidence-grounded response

Existing Systems	SatQuery AI
Single Task	Multi-task
Manual Model Selection	Auto Agent routing
Optical Only	Optical + SAR
Static Analysis	Multi-temporal
No Explainability	Confidence + Evidence

Supported Input Type	Support
Optical Image	Yes
SAR Image	Yes
Optical + SAR Pair	Yes
Bi-temporal Images	Yes
GeoTIFF	Yes
TIFF	Yes
PNG/JPEG	Benchmark only

Schematic diagram of the SatQueryAI

Technologies Used

AI Frameworks



Hugging Face



LangChain



FAISS
(FACEBOOK AI SIMILARITY SEARCH)

Geospatial Libraries

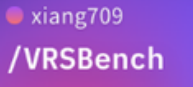


OpenCV **GDAL**

Datasets



sentinel-1
sentinel-2



xiang709
/VRSBench



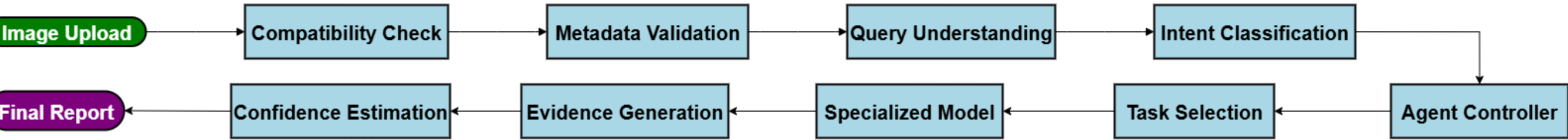
RSVQA

CDVQA

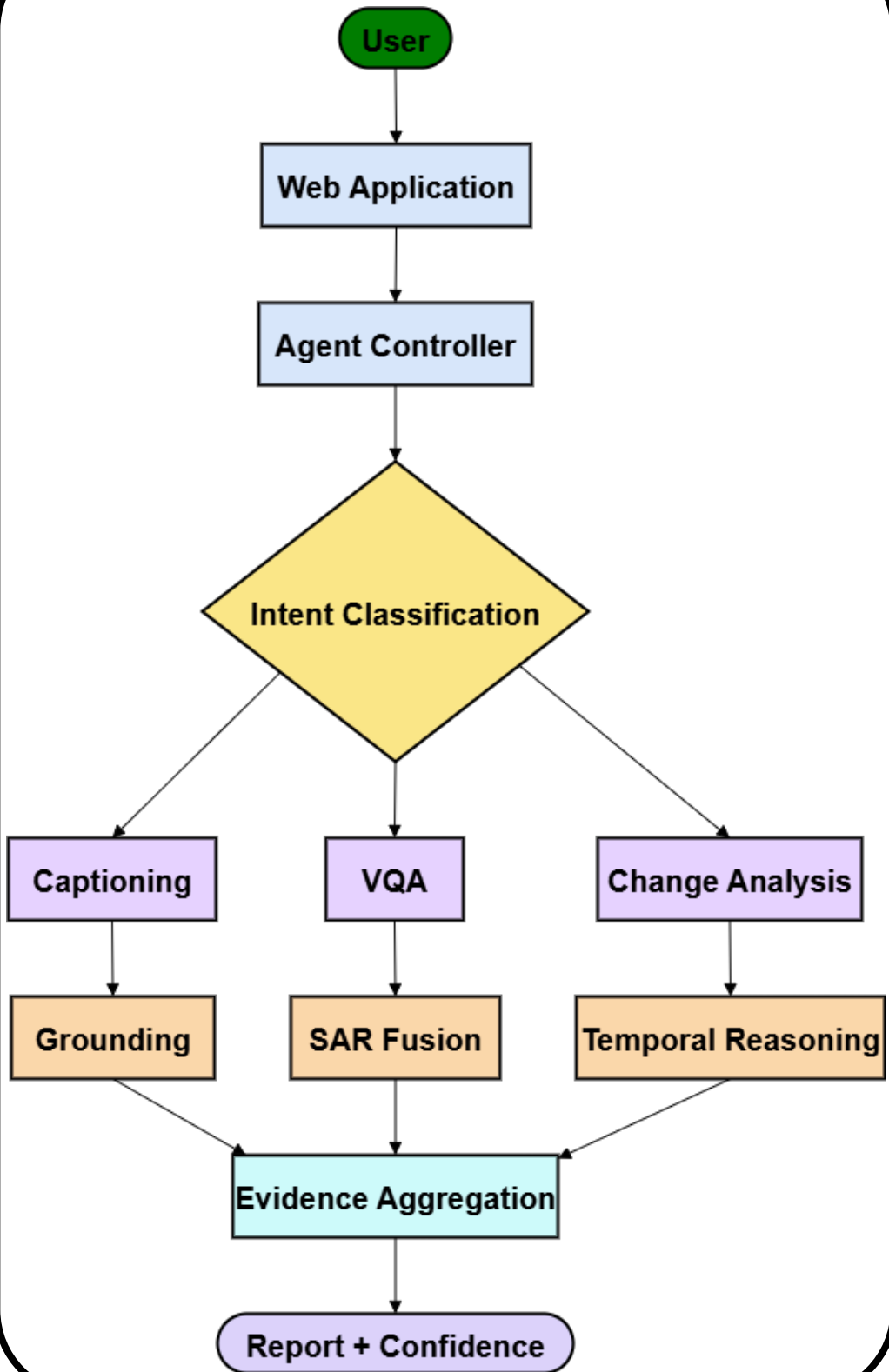
Stage	Description
Input Layer	Accepts GeoTIFF/TIFF, PNG, JPEG, Optical, SAR, and Bi-temporal images
Agent Layer	Understands the query, validates inputs, selects the appropriate workflow
AI Models	VQA, Captioning, Grounding, Change Detection, Optical-SAR Fusion
Output Layer	Generates text answers, change maps, highlighted regions, confidence score, and

Component	Task	Purpose in SatQuery AI
BLIP-2 / LLaVA	Captioning	Generates natural-language descriptions of satellite imagery
GroundingDINO + SAM	Object Grounding	Detects and segments objects from text prompts
RemoteCLIP	Vision-Language Retrieval	Aligns image and text embeddings for semantic understanding
ChangeStar / CDVQA	Change Analysis	Identifies and explains temporal changes between images
Optical-SAR Fusion Network	Multi-modal Fusion	Combines Optical and SAR imagery for robust feature extraction

Methodology



Agentic architecture



BigEarthNet is used to adapt Vision-Language Models to remote sensing imagery.

Technical Feasibility

- Open-source datasets available
- Mature AI ecosystem
- Cloud deployment support
- Scalable modular architecture
- Compatible with ISRO benchmark datasets

Risks & Challenges

- Large satellite image size
- High computational requirements
- SAR image complexity
- Model hallucinations
- Cross-modal alignment

Mitigation Strategy

- Domain adaptation using BigEarthNet
- Retrieval-Augmented Generation
- Confidence estimation
- Dynamic Agent routing
- Parallel multi-model inference
- Execution trace for explainability

Hardware Requirements

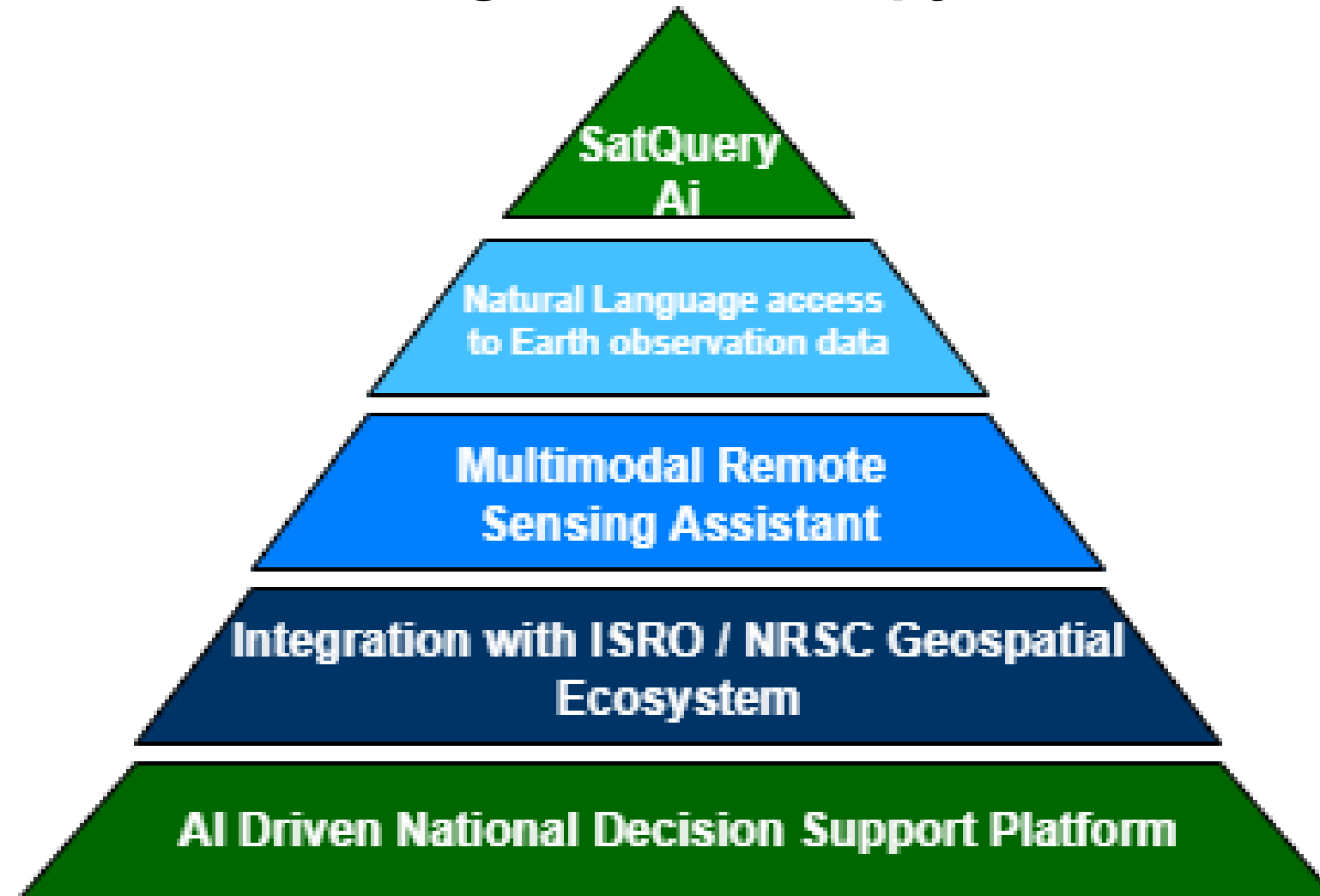
Component	Requirement
Training GPU	NVIDIA RTX 4090 / A100 / TPU
Inference GPU	RTX 3060+
Storage	300–500 GB
Backend	FastAPI
Frontend	React
Deployment	Cloud

Dataset Ecosystem

Dataset	Purpose
BigEarthNet.txt	Remote-sensing domain adaptation & image-text alignment
Sentinel-1 (SAR)	Structural information, all-weather imaging
Sentinel-2 (Optical)	Multispectral land-cover analysis
RSVQA	Visual Question Answering benchmark
VRSBench	Captioning & Grounding evaluation
CDVQA	Multi-temporal change understanding

The proposed solution is technically feasible using open-source datasets, pretrained Vision-Language Models, and a modular agentic architecture. Cloud-ready deployment and scalable hardware requirements enable practical implementation and future expansion.

Long term vision pyramid



Impact of SatQuery AI (Why?)

Current Workflow(Before)	SatQuery AI (After)
GIS expertise required	Ask questions in natural language
Manual model selection	Automatic agentic routing
Single-task analysis	Multi-task reasoning
Slow interpretation	Instant AI-assisted insights
Limited explainability	Confidence scores & execution

Beneficiaries

Disaster Management

- Flood mapping
- Cyclone assessment
- Landslide detection

Defence & Security

- Border monitoring
- Infrastructure intelligence
- Strategic surveillance

Urban Planning

- Infrastructure monitoring
- Smart city planning
- Urban expansion analysis

Forest Monitoring

- Deforestation
- Wildfire assessment
- Biodiversity monitoring

Environmental Monitoring

- Climate studies , Water resource management
- Wetland monitoring

Solution Impact

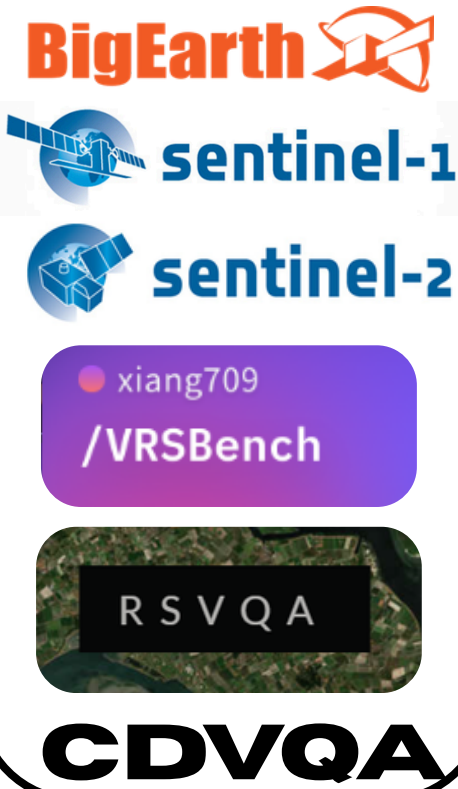
- Natural-language interaction
- Reduced dependency on GIS experts
- Faster remote sensing analysis
- Explainable AI decisions
- Better disaster response
- Improved planning & decision making

Societal Benefits

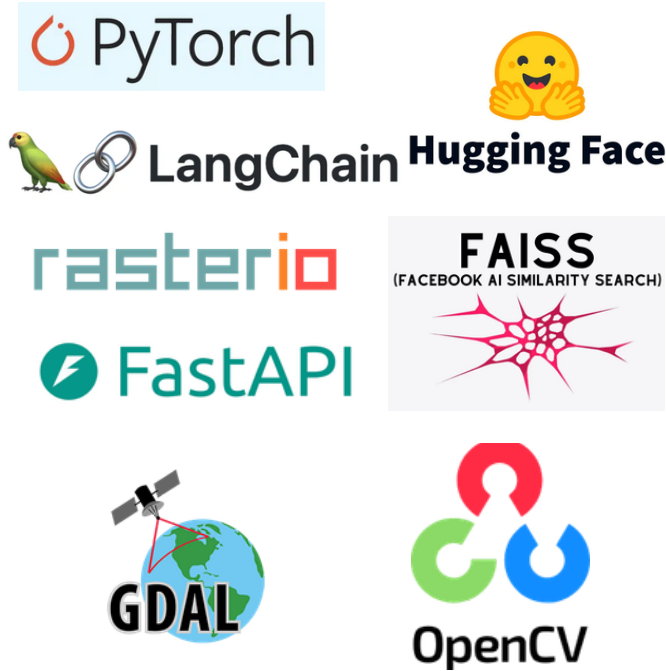
- Faster disaster response
- Reduced analysis time
- Less dependence on specialists
- Lower operational cost
- Better policy decisions
- Democratized Earth observation

SatQuery AI transforms complex multisensor satellite data into evidence-backed insights through an intelligent, agentic, and explainable Artificial intelligence framework—making this Earth observation--accessible, efficient, and impactful.

Datasets



Frameworks



The BigEarthNet archive contains 590 thousand Sentinel-2 image patches with multiple annotations, Examples are given below



Continuous urban fabric, Green urban areas



Non-irrigated arable land, Fruit trees and berry plantations, Pastures.

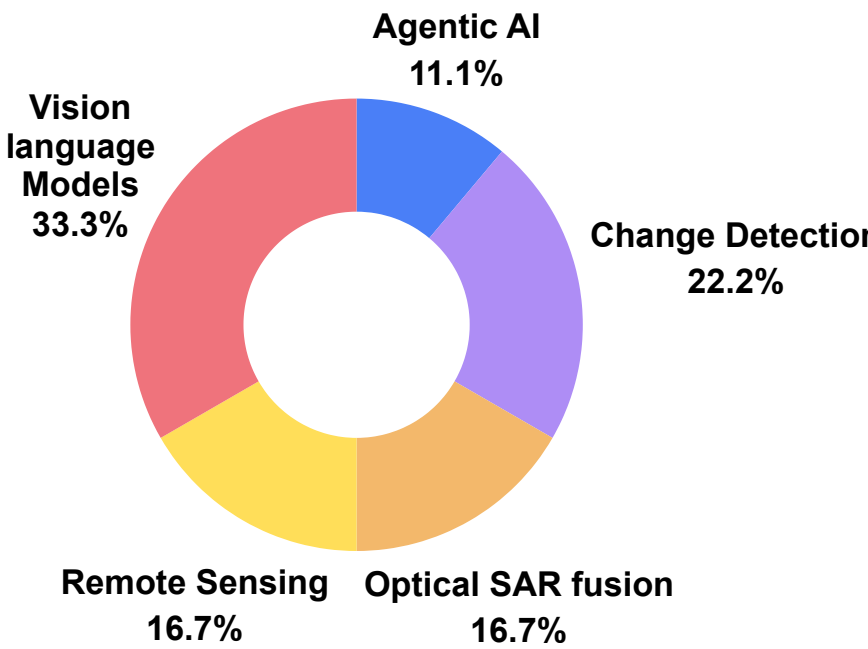


Pastures, water courses, water bodies



Construction sites, Non irrigated arable land, Pastures, Coniferous forest, Inland marshes, Water courses

Literature Review distribution



Resource	Official Link
BigEarthNet with sentinel	BigEarthNet
BigEarthNet.txt	BigEarthNet.txt
VRSBench	VRSBench
RSVQA	RSVQA
Hugging Face	Hugging Face
PyTorch	PyTorch
ESA Sentinel	Copernicus Sentinel Missions

Existing Research	Gap
Single-image VQA	No multimodal reasoning
Captioning	No task orchestration
SAR analysis	No natural-language interface
Change detection	No unified assistant
Generic VLMs	No remote-sensing adaptation

Model	Remote	Optical+SAR	Agentic	Explainable
BLIP-2	NO	NO	NO	NO
LLaVA	NO	NO	NO	NO
GeoChat	YES	NO	NO	LIMITED
RemoteCLIP	YES	Partial	NO	NO
SatQuery AI	YES	YES	YES	YES

Dataset contribution chart

